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CODE:

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# Iris Flower Classification using Naive Bayes

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load the Iris dataset
iris = load_iris()

# Input features
X = iris.data

# Target classes
y = iris.target

# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create Gaussian Naive Bayes model
model = GaussianNB()

# Train the model
model.fit(X_train, y_train)

# Predict the test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Display results
print("IRIS FLOWER CLASSIFICATION USING NAIVE BAYES")
print("=" * 50)

print("\nAccuracy:", round(accuracy * 100, 2), "%")

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=iris.target_names
))

# Classify a new flower
new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_flower)
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print("\nNEW FLOWER DETAILS")
print("-" * 50)
print("Sepal Length:", new_flower[0][0])
print("Sepal Width :", new_flower[0][1])
print("Petal Length:", new_flower[0][2])
print("Petal Width :", new_flower[0][3])

print("\nPredicted Flower:",
      iris.target_names[prediction[0]])

```

OUTPUT :

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IRIS FLOWER CLASSIFICATION USING NAIVE BAYES
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Accuracy: 100.0 %

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

Classification Report:
              precision    recall  f1-score   support

   setosa         1.00        1.00        1.00         10
  versicolor       1.00        1.00        1.00          9
   virginica       1.00        1.00        1.00         11

   accuracy         1.00        1.00        1.00         30
  macro avg         1.00        1.00        1.00         30
 weighted avg         1.00        1.00        1.00         30


NEW FLOWER DETAILS
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Sepal Length: 5.1
Sepal Width : 3.5
Petal Length: 1.4
Petal Width : 0.2

Predicted Flower: setosa

```